

Generation Interconnection System Impact Study Report

For Queue Project Q17-0904

Mickens 138 kV
November 24, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0904. The Interconnection Customer proposes a 75.8 MW battery energy storage facility interconnecting at the Mickens 138 kV substation in the Virginia area. The request was received on December 16, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 270 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,340,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WESTCOAST CELLUFIBRE – LEAVITT 138kV Line 1
Contingency: WESTCOAST CELLUFIBRE – LEAVITT 138kV #2
Loading: 129.5% (Rating: 281 MVA)
DFAX: 8.4%
Pre-project loading: 127.2%
2. Facility: LEAVITT – WESTCOAST CELLUFIBRE 138kV Cir. 2
Contingency: LEAVITT – WESTCOAST CELLUFIBRE 138kV Cir. 1
Loading: 129.5% (Rating: 281 MVA)
DFAX: 8.4%
Pre-project loading: 127.2%
3. Facility: DANC LFGTE FACILITY – ELKO 138kV
Contingency: MIRA VISTA – YARNEN 345kV
Loading: 117.2% (Rating: 228 MVA)
DFAX: 9.3%
Pre-project loading: 114.1%
4. Facility: KEOKUK – KENBRIDGE 138kV
Contingency: MECKLENBURG POWER STATION – KERR DAM 345kV L
Loading: 115.2% (Rating: 320 MVA)
DFAX: 5.2%
Pre-project loading: 114.0%
5. Facility: KLAMATH FALLS – KENBRIDGE 138kV
Contingency: KERR DAM – MECKLENBURG POWER STATION 345kV
Loading: 114.7% (Rating: 326 MVA)
DFAX: 5.2%
Pre-project loading: 113.5%
6. Facility: KLAMATH FALLS – BAUXITE 138kV
Contingency: KERR DAM – MECKLENBURG POWER STATION 345kV
Loading: 114.2% (Rating: 333 MVA)

DFAX: 5.2%
Pre-project loading: 113.0%

7. Facility: RUSSELL FORK – ROBISON PARK 230kV
Contingency: DEKALB SUBSTATION – JONES CHAPEL 500kV
Loading: 113.6% (Rating: 2,461 MVA)
DFAX: 8.4%
Pre-project loading: 113.4%

8. Facility: ELMO – HICKORY 138kV
Contingency: MIRA VISTA – YARNEN 345kV
Loading: 112.1% (Rating: 323 MVA)
DFAX: 14.6%
Pre-project loading: 108.6%

9. Facility: CARROLL – CRITTENDEN 138kV
Contingency: MIRA VISTA – YARNEN 345kV
Loading: 109.8% (Rating: 375 MVA)
DFAX: 15.5%
Pre-project loading: 106.7%

10. Facility: EAST LONGMEADOW – PARSONS TAP 138kV L
Contingency: MCCARTNEY – YARNEN 345kV L
Loading: 109.6% (Rating: 344 MVA)
DFAX: 8.2%
Pre-project loading: 107.8%

11. Facility: CALPINE KING ENERGY – WOLCOTT 138kV
Contingency: MIRA VISTA – YARNEN 345kV
Loading: 106.2% (Rating: 296 MVA)
DFAX: 10.7%
Pre-project loading: 103.5%

12. Facility: PUEBLO XTRA 500 kV/230 kV Transformer Bank
Contingency: PUEBLO XTRA – BOCA RATON SUBSTATION 500kV
Loading: 106.1% (Rating: 1,916 MVA)
DFAX: 7.1%
Pre-project loading: 105.8%

13. Facility: BEAVER – EUFAULA 500kV
Contingency: TREGO – GALION GENERATING STA SUBSTATION 500kV
Loading: 105.8% (Rating: 4,084 MVA)
DFAX: 8.7%
Pre-project loading: 105.6%

14. Facility: CHUCKATUCK – AVRA VALLEY 138kV
Contingency: MIRA VISTA – YARNEN 345kV
Loading: 105.3% (Rating: 568 MVA)
DFAX: 19.8%
Pre-project loading: 102.6%

15. Facility: TILLMAN SUBSTATION – LAKE KINGMAN 138kV line
Contingency: YARNEN SUBSTATION – MIRA VISTA 345kV
Loading: 104.7% (Rating: 301 MVA)
DFAX: 15.5%
Pre-project loading: 100.8%

16. Facility: MCCARTNEY – YARNEN 345kV
Contingency: FYFFE 345 kV/138 kV XFMR
Loading: 104.1% (Rating: 1,656 MVA)
DFAX: 49.2%

Pre-project loading: 101.9%

17. Facility: BRILEY – HONEYVILLE 138kV
Contingency: KERR DAM – MECKENBURG POWER STA 345kV
Loading: 103.9% (Rating: 433 MVA)
DFAX: 8.0%
Pre-project loading: 102.5%
18. Facility: THRASHER – WOLCOTT 138kV
Contingency: YARNEN – MIRA VISTA 345kV Line
Loading: 103.6% (Rating: 287 MVA)
DFAX: 10.7%
Pre-project loading: 100.7%
19. Facility: GALION GENERATING STA – TREGO 500kV
Contingency: TREGO – BEAVER 500kV
Loading: 103.1% (Rating: 2,687 MVA)
DFAX: 5.3%
Pre-project loading: 103.0%
20. Facility: NO MANS LAND – YADKIN 138kV
Contingency: MICKENS – MIRA VISTA 345kV
Loading: 101.6% (Rating: 587 MVA)
DFAX: 32.5%
Pre-project loading: 97.4%
21. Facility: ROANOKE RAPIDS 345 kV/138 kV TX
Contingency: MCCARTNEY – YARNEN 345kV
Loading: 101.4% (Rating: 992 MVA)
DFAX: 24.6%
Pre-project loading: 99.5%
22. Facility: TRAVIS AB – M.L. HIBBARD S.E. STA 138kV
Contingency: YARNEN – MCCARTNEY 345kV Line
Loading: 101.1% (Rating: 329 MVA)
DFAX: 7.8%
Pre-project loading: 99.3%
23. Facility: AVRA VALLEY – WINDSOR DP 138kV
Contingency: YARNEN SUBSTATION – MIRA VISTA 345kV
Loading: 100.8% (Rating: 363 MVA)
DFAX: 9.9%
Pre-project loading: 98.8%
24. Facility: PINE BLUFF W – LAKE KINGMAN 138kV
Contingency: MIRA VISTA – YARNEN 345kV
Loading: 100.3% (Rating: 258 MVA)
DFAX: 17.4%
Pre-project loading: 95.2%
25. Facility: SWAN VLY – MUSKOGEE 500kV
Contingency: RICHLAND – DEKALB 500kV L
Loading: 100.1% (Rating: 3,576 MVA)
DFAX: 5.3%
Pre-project loading: 100.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WESTCOAST CELLUFIBRE – LEAVITT 138kV Line 1	Rebuild line with higher-capacity conductor	\$2,056,000	4% (\$87,000)
LEAVITT – WESTCOAST CELLUFIBRE 138kV Cir. 2	Rebuild line with higher-capacity conductor	\$1,787,000	4% (\$75,000)
ELKO SUBSTATION – DANC LFGTE FACILITY 138kV L	Rebuild line with higher-capacity conductor	\$9,104,000	8% (\$722,000)
KENBRIDGE SUBSTATION – KEOKUK 138kV	Rebuild line with higher-capacity conductor	\$15,579,000	7% (\$1,046,000)
KENBRIDGE – KLAMATH FALLS 138kV Line	Rebuild line with higher-capacity conductor	\$8,727,000	7% (\$586,000)
KLAMATH FALLS – BAUXITE 138kV	Rebuild line with higher-capacity conductor	\$5,408,000	7% (\$363,000)
RUSSELL FORK – ROBISON PARK 230kV	Rebuild line with higher-capacity conductor	\$53,623,000	1% (\$760,000)
ELMO – HICKORY 138kV	Rebuild line with higher-capacity conductor	\$1,622,000	6% (\$89,000)
CARROLL – CRITTENDEN 138kV	Reconductor line	\$1,504,000	9% (\$136,000)
PARSONS TAP – EAST LONGMEADOW 138kV	Reconductor line	\$11,072,000	6% (\$634,000)
CALPINE KING ENERGY – WOLCOTT 138kV	Reconductor line	\$8,143,000	11% (\$912,000)
PUEBLO XTRA 500 kV/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$31,447,000	1% (\$446,000)
BEAVER – EUFAULA 500kV	Reconductor line	\$22,766,000	1% (\$310,000)
CHUCKATUCK – AVRA VALLEY 138kV	Reconductor line	\$2,910,000	8% (\$218,000)
TLLMAN – LAKE KINGMAN 138kV	Reconductor line	\$434,000	9% (\$39,000)
MCCARTNEY – YARNEN 345kV	Reconductor line	\$24,351,000	8% (\$1,861,000)
BRIMLEY – HONEYVILLE 138kV Line	Reconductor line	\$5,773,000	7% (\$387,000)
THRASHER – WOLCOTT 138kV	Reconductor line	\$18,209,000	11% (\$2,039,000)
GALION GENERATING STA – TREGO 500kV	Reconductor line	\$68,072,000	1% (\$946,000)
NO MANS LAND – YADKIN 138kV	Reconductor line	\$1,164,000	9% (\$105,000)
ROANOKE RAPIDS 345 kV/138 kV TX	Replace transformer with higher-rated unit	\$34,605,000	7% (\$2,373,000)
TRAVIS AB – M.L. HIBBARD S.E. STA 138kV	Reconductor line	\$17,596,000	7% (\$1,306,000)
AVRA VALLEY – WINDSOR DP 138kV	Reconductor line	\$1,644,000	9% (\$149,000)
PINE BLUFF W – LAKE KINGMAN SUBSTATION 138kV line	Reconductor line	\$565,000	12% (\$66,000)
SWAN VLY – MUSKOGEE 500kV	Reconductor line	\$37,906,000	2% (\$685,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$16,340,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.